FURTHER MECHANICS / CENTRE OF MASS

Centre of Mass of Non-Uniform Bodies

Mark scheme 10 questions · 115 marks

EXAM BOARDS Edexcel FM2 9FM0/4C

Questions

Question	Marks by part	Marks	Page
Question 1	(a) 3 (b) 4	7	4
Question 2		8	5
Question 3	(a) 3 (b) 5	8	6
Question 4	(a) 3 (b) 4 (c)(i) 4 (c)(ii) 3	14	7
Question 5	(a) 7 (b) 4	11	8
Question 6	(a) 6 (b) 5	11	9
Question 7	(a) 5 (b) 7 (c) 3	15	10
Question 8	(a) 5 (b) 4 (c) 3	12	11
Question 9	(a) 6 (b) 4 (c) 4	14	12
Question 10	(a) 6 (b) 7 (c) 2	15	14
	Total	**115**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

ft	follow through Correct work from the student's own earlier wrong value earns the mark.	**AG**	answer given The answer is printed in the question, so every step to it must be shown.
cao	correct answer only No follow-through from an earlier error.	**cso**	correct solution only No error anywhere in the part, not just in its answer.
oe	or equivalent	**awrt**	answers which round to
soi	seen or implied Earned if written, or if later working shows it was used.	**isw**	ignore subsequent working A correct answer, once reached, isn't spoiled by later working.
SC	special case Credit, as the note says, for work outside the main method.	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$M = \int_0^{3a} k\left(a^2 + x^2\right)\,\mathrm{d}x$ $= k\left[a^2x + \frac{x^3}{3}\right]_0^{3a}$	**M1 A1**	**M1** Sets M equal to the integral of ρ and integrates, at least one power increased by 1. Limits may appear later. **A1** Correct integrated terms.
	$M = k\left(3a^3 + 9a^3\right) = 12ka^3$ $\Rightarrow k = \frac{M}{12a^3}$	**A1 AG**	Limits 0 and $3a$ correctly substituted to reach $M = 12ka^3$ oe, then the printed result without error.
		[3]	
1(b)	$M\bar{x} = \int_0^{3a} k\left(a^2x + x^3\right)\,\mathrm{d}x$ $= k\left[\frac{a^2x^2}{2} + \frac{x^4}{4}\right]_0^{3a}$	**M1**	Integrates $x\rho$, at least one power increased by 1. Condone coefficient slips; limits may appear later.
	$= \frac{99ka^4}{4}$	**A1**	oe, or $\frac{33Ma}{16}$ with k substituted.
	$\bar{x} = \text{“}\frac{99ka^4}{4}\text{”} \div 12ka^3 = \ldots$	**dM1**	Divides their moment by $12ka^3$, or by M with $k = \frac{M}{12a^3}$ substituted.
	$\bar{x} = \frac{33a}{16}$	**A1**	oe, such as $2.0625a$.
		[4]	
Total for Question 1		**7**	

Question	Scheme	Marks	Notes
2	$M = \int_0^a k\left(1 + e^{-x/a}\right) \mathrm{d}x$ $= k\left[x - ae^{-x/a}\right]_0^a$	**M1**	Integrates the density to the form $px + qe^{-x/a}$. Condone sign and coefficient slips.
	$M = \frac{ka(2e-1)}{e}$	**A1**	oe, such as $k\left(2a - \frac{a}{e}\right)$.
	$M\bar{x} = \int_0^a kx\left(1 + e^{-x/a}\right) \mathrm{d}x$	**M1**	Writes the moment about A as the integral of x times the density. Limits may be implied by later working.
	$\int xe^{-x/a} \mathrm{d}x = -axe^{-x/a} + \int ae^{-x/a} \mathrm{d}x$ $= -axe^{-x/a} - a^2e^{-x/a}$	**dM1**	Integrates $xe^{-x/a}$ by parts, or any valid method, to the form $pxe^{-x/a} + qe^{-x/a}$. Condone sign and coefficient slips.
	$M\bar{x} = k\left[\frac{x^2}{2} - axe^{-x/a} - a^2e^{-x/a}\right]_0^a$	**A1**	All integrated terms correct, oe.
	$M\bar{x} = \frac{ka^2(3e-4)}{2e}$	**A1**	oe, such as $ka^2\left(\frac{3}{2} - \frac{2}{e}\right)$.
	$\bar{x} = \text{“}\frac{ka^2(3e-4)}{2e}\text{”} \times \text{“}\frac{e}{ka(2e-1)}\text{”}$	**dM1**	Divides their moment by their mass, both found by integration.
	$\bar{x} = \frac{(3e-4)a}{2(2e-1)}$	**A1 AG**	Correct mass and moment leading without error to the printed result; combined simplification accepted.
	Total for Question 2	**8**	

Question	Scheme	Marks	Notes
3(a)	$M = \int_0^a kx\cos\left(\frac{\pi x}{2a}\right)\mathrm{d}x$ $= \frac{2akx}{\pi}\sin\left(\frac{\pi x}{2a}\right) - \int \frac{2ak}{\pi}\sin\left(\frac{\pi x}{2a}\right)\mathrm{d}x$	**M1**	Integrates $\int_0^a \rho\,\mathrm{d}x$ by parts to the form $px\sin\left(\frac{\pi x}{2a}\right) \pm q\int \sin\left(\frac{\pi x}{2a}\right)\mathrm{d}x$, or the tabular method's two-term result. Condone sign and coefficient slips.
	$M = k\left[\frac{2ax}{\pi}\sin\left(\frac{\pi x}{2a}\right) + \frac{4a^2}{\pi^2}\cos\left(\frac{\pi x}{2a}\right)\right]_0^a$	**A1**	Correct integrated terms; k may appear inside or outside.
	$M = k\left(\frac{2a^2}{\pi} - \frac{4a^2}{\pi^2}\right)$ $= \frac{2ka^2(\pi-2)}{\pi^2}$	**A1 AG**	Both limits correctly substituted, including the nonzero value at $x = 0$, leading without error to the printed mass; combined steps accepted.
		[3]	
3(b)	$\int x^2\cos\left(\frac{\pi x}{2a}\right)\mathrm{d}x$ $= \frac{2ax^2}{\pi}\sin\left(\frac{\pi x}{2a}\right) - \frac{4a}{\pi}\int x\sin\left(\frac{\pi x}{2a}\right)\mathrm{d}x$	**M1**	Uses $\int_0^a x\rho\,\mathrm{d}x$ and integrates by parts to the form $px^2\sin\left(\frac{\pi x}{2a}\right) \pm q\int x\sin\left(\frac{\pi x}{2a}\right)\mathrm{d}x$, or the tabular method's three-term result. Condone sign and coefficient slips.
	$\int x\sin\left(\frac{\pi x}{2a}\right)\mathrm{d}x$ $= -\frac{2ax}{\pi}\cos\left(\frac{\pi x}{2a}\right) + \frac{4a^2}{\pi^2}\sin\left(\frac{\pi x}{2a}\right)$	**dM1**	Integrates by parts again, reaching the form $px\cos\left(\frac{\pi x}{2a}\right) + q\sin\left(\frac{\pi x}{2a}\right)$, or the tabular result with these terms. Condone sign and coefficient slips.
	$M\bar{x} = k\left(\frac{2a^3}{\pi} - \frac{16a^3}{\pi^3}\right)\ \left[= \frac{2ka^3(\pi^2-8)}{\pi^3}\right]$	**A1**	Correct moment, oe; need not be simplified.
	$\bar{x} = \text{“}\frac{2ka^3(\pi^2-8)}{\pi^3}\text{”} \div \frac{2ka^2(\pi-2)}{\pi^2}$	**dM1**	Divides their moment by the mass from (a).
	$\bar{x} = \frac{a(\pi^2-8)}{\pi(\pi-2)}$	**A1**	Any exact equivalent.
		[5]	
	Total for Question 3	**8**	

Question	Scheme	Marks	Notes
4(a)	$(m_C =)\ \pi\int_0^3 \lambda(12-x^2)\times 2^2\,\mathrm{d}x$ $= 4\pi\lambda\left[12x-\frac{x^3}{3}\right]_0^3$	**M1 A1**	**M1** Uses $\pi\int\rho y^2\,\mathrm{d}x$ with $y=2$ and integrates, raising at least one power by 1. Limits need not be applied. **A1** Correct integrated expression.
	$4\pi\lambda(36-9)=108\pi\lambda$	**A1 AG**	Correct limits substituted, leading without error to $108\pi\lambda$.
		[3]	
4(b)	$(m_C\bar{x}_C =)\ 4\pi\lambda\int_0^3(12x-x^3)\,\mathrm{d}x$ $= 4\pi\lambda\left[6x^2-\frac{x^4}{4}\right]_0^3$	**M1**	Uses $\pi\int\rho xy^2\,\mathrm{d}x$ with $y=2$ and integrates, raising at least one power by 1.
	$m_C\bar{x}_C = 135\pi\lambda$	**A1**	
	$108\pi\lambda\,\bar{x}_C =$ "$135\pi\lambda$" $\Rightarrow \bar{x}_C = \ldots$	**dM1**	Divides their moment integral's value by $108\pi\lambda$.
	$\bar{x}_C = \frac{135}{108} = 1.25$	**A1 AG**	Correct $\frac{135\pi\lambda}{108\pi\lambda}$ oe leading to 1.25, with no errors in the part.
		[4]	
4(c)(i)	$2\lambda\times\frac{1}{3}\pi\times 3^2\times 4 = 24\pi\lambda$	**B1**	Cone's mass $24\pi\lambda$ oe.
	$3+\frac{1}{4}\times 4 = 4$	**B1**	Cone's centre of mass 4 cm from O, or 1 cm from A used correctly.
	"$132\pi\lambda$" $\bar{x} = 108\pi\lambda\times 1.25$ $+$ "$24\pi\lambda$" $\times$ "4"	**M1**	Forms a moments equation about any point on the axis, with both pieces' masses and total mass their sum. Condone a slip in one value.
	$\bar{x} = \frac{231}{132} = 1.75$	**A1 AG**	Correct equation leading without error to 1.75 from O.
		[4]	
4(c)(ii)	Centre of mass vertically above the lowest point of the end containing O: $\tan\alpha^\circ = \frac{2}{1.75}\ \left[=\frac{8}{7}\right]$	**M1 A1**	**M1** Puts the centre of mass above the lowest point of the end containing O and forms a ratio for α from that end's radius and the distance from O. Condone the ratio inverted. **A1** $\frac{2}{1.75}$ oe.
	$\alpha = 48.8$	**A1**	awrt 48.8.
		[3]	
	Total for Question 4	**14**	

Question	Scheme	Marks	Notes
5(a)	$y = \frac{x}{2}$	**B1**	Radius at distance x from V, oe.
	$M = \pi \int_0^{2a} k\left(3 - \frac{x}{a}\right) \frac{x^2}{4}\,\mathrm{d}x$ $= \frac{\pi k}{4}\left[x^3 - \frac{x^4}{4a}\right]_0^{2a}$	**M1**	Integrates $\pi\rho y^2$ with their y in terms of x, at least one power increased by 1. Condone sign and coefficient slips.
	$= \frac{\pi k}{4}\left(8a^3 - 4a^3\right) = \pi k a^3$	**A1**	Printed mass from correct integration and limits.
	$M\bar{x} = \pi \int_0^{2a} k\left(3 - \frac{x}{a}\right) \frac{x^3}{4}\,\mathrm{d}x$ $= \frac{\pi k}{4}\left[\frac{3x^4}{4} - \frac{x^5}{5a}\right]_0^{2a}$	**M1**	Integrates $\pi\rho x y^2$ with their y in terms of x, at least one power increased by 1. Condone sign and coefficient slips.
	$= \frac{7}{5}\pi k a^4$	**A1**	oe
	$\bar{x} = \frac{\frac{7}{5}\pi k a^4}{\pi k a^3} = \frac{7}{5}a$	**dM1** **A1 AG**	**dM1** Divides their moment by their mass or $\pi k a^3$. **A1 AG** Correct moment and mass leading without error to the printed result.
		[7]	
5(b)	$(M+m)\bar{x}_G = \frac{7}{5}Ma + 2ma$ $(M+m)\bar{y}_G = ma$	**M1 A1**	**M1** Forms moment equations for both coordinates of the combined centre of mass, or takes moments about V for the hanging body, with both masses. Condone slips. **A1** Both correct, oe, or a correct moment equation about V.
	$\tan\theta = \frac{\text{“}ma\text{”}}{\text{“}\frac{7}{5}Ma + 2ma\text{”}} = \frac{1}{4}$ $\Rightarrow m = \ldots$	**dM1**	Sets their perpendicular-to-axial offset ratio of the centre of mass equal to $\frac{1}{4}$, or uses $\tan\theta = \frac{1}{4}$ in their moment equation, and solves for m. Condone the ratio inverted.
	$m = \frac{7}{10}M$	**A1**	oe
		[4]	
Total for Question 5		**11**	

Question	Scheme	Marks	Notes
6(a)	$(m_S =)\ \pi\int_0^4 \lambda(6-x)(4x)\,\mathrm{d}x$ $= 4\pi\lambda\left[3x^2 - \frac{x^3}{3}\right]_0^4$	**M1**	Uses $\int(6-x)y^2\,\mathrm{d}x$ with $y^2 = 4x$ and integrates, power of at least one term increased by 1.
	$m_S = \frac{320\pi\lambda}{3}$	**A1**	oe. Condone π, λ or both omitted consistently from mass and moment.
	$(m_S\bar{x}_S =)\ \pi\int_0^4 \lambda(6-x)x(4x)\,\mathrm{d}x$ $= 4\pi\lambda\left[2x^3 - \frac{x^4}{4}\right]_0^4$	**M1**	Uses the mass integrand with an extra factor x and integrates, power of at least one term increased by 1.
	$m_S\bar{x}_S = 256\pi\lambda$	**A1**	oe.
	$\bar{x}_S =$ "$256\pi\lambda$" $\div$ "$\frac{320\pi\lambda}{3}$" $= \ldots$	**dM1**	Divides their moment by their mass.
	$\bar{x}_S = 2.4\,\mathrm{cm}$	**A1**	
		[6]	
6(b)	Centre of mass of toy at centre of common face: $\bar{x} = 4$	**B1**	Stated or used, as $\bar{x} = 4$ or zero resultant moment about the common centre. No justification needed.
	Hemisphere's centre of mass at $x = 4 + 1.5 = 5.5$	**B1**	1.5 cm from the face, away from O, oe for their origin.
	$\left("\frac{320\pi\lambda}{3}" + \frac{128\pi\rho}{3}\right) \times 4$ $= "\frac{320\pi\lambda}{3}" \times "2.4" + \frac{128\pi\rho}{3} \times 5.5$	**M1 A1**	**M1** Forms a moments equation using both pieces' masses and positions, with the toy's centre of mass at the common centre. Condone sign slips. **A1** Correct equation oe, such as $\frac{320\pi\lambda}{3} \times 1.6 = \frac{128\pi\rho}{3} \times 1.5$.
	$\lambda = \frac{3\rho}{8}$	**A1**	oe, such as $\lambda = 0.375\rho$.
		[5]	
Total for Question 6		**11**	

Question	Scheme	Marks	Notes
7(a)	$OA : x = 2y$, strip length $4a - 2y$	**B1**	oe, soi in the integrand.
	$M = \int_0^{2a} k\left(1 + \frac{y}{a}\right)$ "$(4a - 2y)$" $\mathrm{d}y$ $= k\left[4ay + y^2 - \frac{2y^3}{3a}\right]_0^{2a}$	**M1 A1**	**M1** Integrates ρ times their strip length with respect to y; at least one power increased by 1. Condone slips in expanding. **A1** Correct integrated expression, any form.
	$M = k\left(8a^2 + 4a^2 - \frac{16a^2}{3}\right)$ $= \frac{20}{3}ka^2$	**dM1** **A1 AG**	**dM1** Substitutes $y = 2a$ into their integrated expression. **A1 AG** Correct substituted expression leading without error to $M = \frac{20}{3}ka^2$.
		[5]	
7(b)	$x = \frac{2y+4a}{2} = y + 2a$	**B1**	Midpoint of the strip, oe; in another valid moment integral, its distance factor.
	$M\bar{x} = \int_0^{2a}$ "$(y + 2a)$" $k\left(1 + \frac{y}{a}\right)$ "$(4a - 2y)$" $\mathrm{d}y$ $= k\left[8a^2y + 4ay^2 - \frac{2y^3}{3} - \frac{y^4}{2a}\right]_0^{2a}$	**M1**	Integrates their strip x times ρ times their strip length with respect to y, or any valid moment integral; at least one power increased by 1. Condone slips in expanding.
	$M\bar{x} = \frac{56}{3}ka^3$	**A1**	
	$\bar{x} = \frac{56}{3}ka^3 \div \frac{20}{3}ka^2 = \frac{14a}{5}$	**A1**	oe, such as $2.8a$.
	$M\bar{y} = \int_0^{2a} y\,k\left(1 + \frac{y}{a}\right)$ "$(4a - 2y)$" $\mathrm{d}y$ $= k\left[2ay^2 + \frac{2y^3}{3} - \frac{y^4}{2a}\right]_0^{2a}$	**M1**	Integrates y times ρ times their strip length with respect to y, or any valid moment integral; at least one power increased by 1. Condone slips in expanding.
	$M\bar{y} = \frac{16}{3}ka^3$	**A1**	
	$\bar{y} = \frac{16}{3}ka^3 \div \frac{20}{3}ka^2 = \frac{4a}{5}$	**A1**	oe, such as $0.8a$.
		[7]	
7(c)	G vertically below A: $\tan\theta = \frac{4a - \text{“}\frac{14a}{5}\text{”}}{2a - \text{“}\frac{4a}{5}\text{”}}$ $[= 1]$	**M1 A1ft**	**M1** Takes AG vertical and forms a trigonometric ratio from the offsets of their G from A, or any valid trigonometry in triangle ABG. Condone the ratio inverted. **A1ft** Correct ratio the right way up, ft their $\bar{x}$ and $\bar{y}$.
	$\theta = 45°$	**A1**	cso; $\frac{\pi}{4}$ accepted.
		[3]	
	Total for Question 7	**15**	

Question	Scheme	Marks	Notes
8(a)	$\delta m = k(2a-r) \times \frac{2\pi}{3} r\,\delta r$	**B1**	Correct mass of a thin arc, oe; may be seen in the integrand.
	$M = \frac{2\pi k}{3}\int_0^a (2ar - r^2)\,\mathrm{d}r$ $= \frac{2\pi k}{3}\left[ar^2 - \frac{r^3}{3}\right]_0^a$	**M1 A1**	**M1** Integrates their mass element with respect to r, the power of at least one term increased by 1. **A1** Correct integrated terms; limits need not be seen.
	$M = \frac{2\pi k}{3} \times \frac{2a^3}{3}$ $= \frac{4\pi a^3}{9} \times \frac{3\lambda}{\pi a^3} = \ldots$	**dM1**	Applies limits 0 and a and substitutes $k = \frac{3\lambda}{\pi a^3}$, in either order.
	$M = \frac{4\lambda}{3}$	**A1**	
		[5]	
8(b)	$M\bar{x} = \int_0^a \frac{3\sqrt{3}\,r}{2\pi} \times k(2a-r)\frac{2\pi}{3} r\,\mathrm{d}r$ $= \sqrt{3}\,k\left[\frac{2ar^3}{3} - \frac{r^4}{4}\right]_0^a$	**M1**	Forms a moment integral of their arc mass times the given arc distance $\frac{r\sin\alpha}{\alpha}$, or any valid moment integral, and integrates. Condone a wrong α.
	$M\bar{x} = \frac{5\sqrt{3}\,ka^4}{12}$	**A1**	oe, such as $\frac{5\sqrt{3}\,\lambda a}{4\pi}$.
	$\bar{x} = \text{“}\frac{5\sqrt{3}\,ka^4}{12}\text{”} \div \text{“}\frac{4\pi k a^3}{9}\text{”} = \ldots$	**dM1**	Divides their moment by their mass from (a), both in k or both in λ.
	$\bar{x} = \frac{45\sqrt{3}\,a}{48\pi} = \frac{15\sqrt{3}\,a}{16\pi}$	**A1 AG**	Correct moment and mass leading without error to the printed result.
		[4]	
8(c)	Uses AG vertical, with $\angle AOG = \frac{\pi}{3}$: $\tan\theta = \frac{\bar{x}\sin\frac{\pi}{3}}{a - \bar{x}\cos\frac{\pi}{3}}$	**M1**	Uses AG vertical, forming a trigonometric equation for $\angle OAG$ in triangle OAG from $\bar{x}$, a and $\frac{\pi}{3}$. Condone the ratio inverted.
	$\tan\theta = \frac{0.4476a}{0.7416a} \approx 0.604$	**A1**	Correct value of $\tan\theta$, or another correct trigonometric ratio for $\angle OAG$.
	$\theta = 31.1°$	**A1**	awrt 31.1°.
		[3]	
	Total for Question 8	**12**	

Question	Scheme	Marks	Notes
9(a)	$m_D = \pi \int_{3a}^{5a} \frac{\lambda}{x^2}\left(25a^2 - x^2\right) \mathrm{d}x$ $= \pi\lambda\left[-\frac{25a^2}{x} - x\right]_{3a}^{5a}$	**M1**	Forms $\int \rho y^2 \,\mathrm{d}x$ with $y^2 = 25a^2 - x^2$ and integrates at least one term correctly.
	$m_D = \frac{4\pi\lambda a}{3}$	**A1**	oe. Condone π, λ or both omitted consistently from mass and moment.
	$m_D\bar{x}_D = \pi\lambda\int_{3a}^{5a}\left(\frac{25a^2}{x} - x\right)\mathrm{d}x$ $= \pi\lambda\left[25a^2\ln x - \frac{1}{2}x^2\right]_{3a}^{5a}$	**M1**	Forms the moment integral with the extra factor x and integrates to the form $p\ln x + qx^2$. Condone coefficient slips.
	$m_D\bar{x}_D = \pi\lambda a^2\left(25\ln\frac{5}{3} - 8\right)$	**A1**	oe, such as $\pi\lambda\left(25a^2\ln\frac{5}{3} - 8a^2\right)$.
	$\bar{x}_D = \frac{\text{“}\pi\lambda a^2\left(25\ln\frac{5}{3}-8\right)\text{”}}{\text{“}\frac{4}{3}\pi\lambda a\text{”}}$	**dM1**	Divides their moment by their mass.
	$\bar{x}_D = \left(\frac{75}{4}\ln\frac{5}{3} - 6\right)a$ $\bar{x}_D - 3a = \left(\frac{75}{4}\ln\frac{5}{3} - 9\right)a$	**A1 AG**	Correct mass and moment, divided and measured from $x = 3a$, reaching the printed result without error.
		[6]	
9(b)	Cylinder at $x = 0$, dome at $x = \left(\frac{75}{4}\ln\frac{5}{3} - 6\right)a$	**B1**	Both centres of mass on the axis from one consistent origin, the dome's from part (a).
	$(k+4)W\bar{x} = kW \times 0$ $+ 4W\left(\frac{75}{4}\ln\frac{5}{3} - 6\right)a$	**M1 A1**	**M1** Forms a moments equation with the total and both pieces, about any point on the axis. Condone one sign slip. **A1** Correct equation, oe, with weights or masses in the ratio $k : 4$.
	$\bar{x} = \frac{\left(75\ln\frac{5}{3}-24\right)a}{k+4}$ $d = \lvert\bar{x} - 3a\rvert = \frac{\left\lvert 3k+36-75\ln\frac{5}{3}\right\rvert}{k+4}a$	**A1 AG**	Correct equation leading without error to the printed result. Condone modulus signs not used.
		[4]	
9(c)	$\tan\beta = \frac{4a}{\bar{x}+3a} = \frac{4}{5}$	**M1**	Uses a right-angled triangle with the radius $4a$ and the distance of G from the free face. Condone the ratio inverted.
	$\bar{x} + 3a = 5a \Rightarrow \bar{x} = 2a$	**A1**	G at $5a$ from the free face, or $d = a$, oe in any origin.

Question 9 continued

Question	Scheme	Marks	Notes
	$\frac{75\ln\frac{5}{3}-24}{k+4} = \text{“2”} \Rightarrow k = \ldots$	**M1**	Equates their position of G with part (b)'s expression in the same origin, or their d with part (b)'s d, and solves for k.
	$k = \frac{75}{2}\ln\frac{5}{3} - 16$	**A1**	Any exact equivalent. A negative value from the modulus must be rejected.
		[4]	
	Total for Question 9	**14**	

Question	Scheme	Marks	Notes
10(a)	$W = \int_0^l \lambda(l+kx)\,\mathrm{d}x$ $= \lambda\left[lx + \frac{1}{2}kx^2\right]_0^l$	**M1**	Integrates $\lambda(l+kx)$ from 0 to l, power of at least one term increased by 1. Condone λ omitted consistently.
	$W = \frac{1}{2}\lambda l^2(2+k)$	**A1**	oe, with λ or consistently without it.
	$W\bar{x} = \int_0^l \lambda x(l+kx)\,\mathrm{d}x$ $= \lambda\left[\frac{1}{2}lx^2 + \frac{1}{3}kx^3\right]_0^l$	**M1**	Integrates x times their weight per unit length from 0 to l, power of at least one term increased by 1.
	$W\bar{x} = \frac{1}{6}\lambda l^3(3+2k)$	**A1**	oe, with λ or consistently without it.
	$\bar{x} = \frac{\frac{1}{6}\lambda l^3(3+2k)}{\frac{1}{2}\lambda l^2(2+k)}$ $= \frac{(3+2k)\,l}{3(2+k)}$	**dM1** **A1 AG**	**dM1** Divides their moment by their weight. **A1 AG** Correct weight and moment leading without error to the printed distance.
		[6]	
10(b)	$\bar{x} = \frac{(3+6)\,l}{3\times 5} = \frac{3}{5}l$	**B1**	
	$S \times \frac{4}{5}l = W \times$ "$\frac{3}{5}l$" $\cos\theta$ $\Rightarrow S = \frac{3}{4}W\cos\theta$	**M1 A1**	**M1** Takes moments about A, or another point, with all relevant terms, the weight at their $\frac{3}{5}l$ and resolved. Condone sign slips and sin/cos confusion.
	$F = S\sin\theta = \frac{3}{4}W\sin\theta\cos\theta$ $R = W - S\cos\theta = W - \frac{3}{4}W\cos^2\theta$	**M1 A1**	**M1** Resolves horizontally and vertically, their S resolved in both. Condone sign slips and sin/cos confusion. **A1** Both correct, oe.
	"$\frac{3}{4}W\sin\theta\cos\theta$" $\leq \mu\left($"$W - \frac{3}{4}W\cos^2\theta$"$\right)$	**M1**	Uses $F \leq \mu R$, or $F = \mu R$ for limiting equilibrium, with their F and R in terms of W and θ.
	$\mu \geq \frac{3\sin\theta\cos\theta}{4-3\cos^2\theta}$ $= \frac{3\tan\theta}{1+4\tan^2\theta}$	**A1 AG**	Correct F and R leading without error to the printed result with $\geq$; from $F = \mu R$, the inequality must be stated.
		[7]	
10(c)	Greatest value of $\frac{3t}{1+4t^2}$, $t = \tan\theta > 0$: $(2t-1)^2 \geq 0 \Rightarrow \frac{3t}{1+4t^2} \leq \ldots$	**M1**	Maximises the expression from (b) by any method, such as an inequality, a derivative set to zero or a discriminant, leading to a value.

Question 10 continued

Question	Scheme	Marks	Notes
	$\mu = \frac{3}{4}$	**A1**	
		[2]	
	Total for Question 10	**15**	